ABDULLAH ALSUMAIRI

Environmental Specialist | EIA & Compliance | GIS & Remote Sensing
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PROFESSIONAL PROFILE

Environmental specialist with an M.Sc. in Environmental Science (First-Class Honors) from King Saud University and experience across environmental consulting, wildlife conservation, and independent geospatial work. Prepared EIA documents aligned with NCEC requirements, led biodiversity surveys and habitat suitability modeling at the National Center for Wildlife, and now provides freelance GIS, remote sensing, and spatial modeling consultancy. Co-author of a peer-reviewed article in *BMC Ecology and Evolution* (2026). Bridges regulatory compliance with data-driven environmental analysis.

EDUCATIONAL QUALIFICATIONS

M.Sc. Environmental Science, King Saud University, Riyadh | 2024 – July 2026 | GPA 4.81 / 5, First-Class Honors
B.Sc. Biology, Shaqra University, Shaqra | 2019 – 2024 | GPA 4.3 / 5, Second-Class Honors

PROFESSIONAL EXPERIENCE

Independent GIS & Environmental Modeling Consultant | Freelance | July 2026 – Present | Riyadh, Saudi Arabia

- Provide GIS, remote sensing, and spatial modeling consultancy to multiple organizations on mapping and environmental analysis projects.
- Build national-scale 30 m spatial datasets and reproducible R/Python workflows for terrain, landscape, and ecosystem classification across Saudi Arabia.
- Analyze Sentinel-2 and MODIS time series (STAC retrieval, harmonic trend analysis) for vegetation monitoring and change detection, delivering publication-quality maps and technical summaries.

Environmental Specialist | National Center for Wildlife (NCW), Skill Development Program | September 2025 – March 2026 | Riyadh, Saudi Arabia

- Led biodiversity surveys and ecosystem classification activities in protected areas to support wildlife conservation planning.
- Developed MaxEnt habitat suitability models and QGIS biodiversity maps; co-authored the resulting peer-reviewed study with the Ecosystems Department.
- Analyzed environmental field data in R and translated findings into technical reports and stakeholder presentations.
- Conducted pre-fieldwork risk assessments to identify hazards and ensure safe field operations.

Environmental Specialist | Environmental Way Consulting Office | April 2025 – August 2025 | Riyadh, Saudi Arabia

- Conducted environmental site assessments to evaluate pollution conditions and collect data for regulatory submissions.
- Prepared EIA documents and site assessment reports in line with National Center for Environmental Compliance (NCEC) requirements and national standards.
- Reviewed applicable environmental regulations to ensure project compliance.

SELECTED PROJECTS

Potential Plant Communities of Saudi Arabia: A Literature-Informed National Atlas | 2026
Synthesized 30 published Saudi vegetation studies (240 source communities) into 26 evidence-graded potential community types, mapped on a 250 m national grid from climate, terrain, substrate and satellite indicators, with a companion knowledge-gap map.

Saudi Arabia National Landscape Units Mapping | 2026
Produced a Kingdom-wide map of 11 landscape units (mountains, harrat, wadis, rawdhat, sabkha, sand seas, and others) in R and QGIS, processing 30 m Copernicus DEM across 870 tiles with geomorphon and multiscale TPI analysis within a GEOSA-based boundary.

Environmental Card System (ECS) | Envirothon 2026, Environment Fund — Co-developed an AI-based platform that scores facility environmental performance; team selected among the top teams from 300+ ideas.

Saudi Native Flora | saudi-native-flora.vercel.app — Designed and built a bilingual visual field guide to 59 native Saudi plant species, verified against WCVP, with GBIF occurrence maps and cited references.

RESEARCH & PUBLICATIONS

PEER-REVIEWED PUBLICATION

Al-Hashimi A, Darwish M, Al-farraj NS, Al-otaibi N, **Al-otaibi A**, Al-majed M, Al-amri SS, Boug A (2026). Escarpment refugia under threat: climate-driven redistribution modeling of four endemic succulent species in Saudi Arabia. *BMC Ecology and Evolution*. doi.org/10.1186/s12862-026-02582-0 (Abdullah Alsumairi, published as Al-Otaibi, A.)

M.SC. THESIS, KING SAUD UNIVERSITY (2026)

MaxEnt-Based Assessment of Current Habitat Suitability and Environmental Associations for Five Narrow-Range Endemic Plant Species in Saudi Arabia.

MANUSCRIPT IN PREPARATION

National-Scale Vegetation Greenness Dynamics in Saudi Arabia, 2016–2025: A MODIS NDVI and ERA5 Assessment of Change over the Saudi Green Initiative Period (2026).

RESEARCH PROPOSAL

Geographic distribution of liver disease and agricultural land cover across Saudi regions. External Co-Investigator (study design, GIS and environmental methodology), King Faisal Specialist Hospital & Research Centre (KFSHRC), 2026.

TECHNICAL SKILLS

GIS & Remote Sensing: QGIS, Google Earth Engine, Sentinel-2 & MODIS time-series analysis, STAC-based image retrieval, geomorphometric analysis (geomorphons, TPI, terrain derivatives), hydrological modeling (flow routing, watershed delineation, TWI, HAND), digital soil mapping, MCDA (AHP), WhiteboxTools. Datasets: WorldClim, CHELSA, ERA5, CHIRPS, SoilGrids, Copernicus DEM, ESA WorldCover, GBIF.

Ecological Modeling: Species distribution modeling (MaxEnt, maxnet), CMIP6 projections and range-change analysis, habitat mapping, ecosystem classification (IUCN GET).

Data Analysis & Programming: R (terra, sf, gstat, ranger, usdm), Python (rasterio, geopandas, xarray), geostatistics (regression kriging, downscaling), trend analysis (Mann–Kendall, Theil–Sen, harmonic regression), VIF screening, accuracy assessment (Kappa), Excel, Power BI, web mapping (Next.js, Vercel).

Field Work: Biodiversity surveys, ecosystem classification, UAV/drone operations. **Frameworks & Standards:** EIA, ISO 14001, IUCN Red List, KBA, Ramsar, NCEC, NCW, NCVC.

CERTIFICATIONS & LICENSES

- Remote Pilot Certificate – Open Category, General Authority of Civil Aviation (GACA), 2025
- IUCN Red List of Ecosystems, IUCN Academy, 2025
- Identifying and Delineating Key Biodiversity Areas (KBAs), IUCN / CBD, 2025
- Introduction to Mapping Biodiversity Priorities, IUCN Academy, 2025
- Introductory Course on the Ramsar Convention on Wetlands, 2025
- Introduction to Plant Classification and Monitoring, National Center for Wildlife, 2025
- ISO 14001 Environmental Management System (EMS), IIRSM Certified
- ISO 45001 Occupational Health and Safety Management System, IIRSM Certified
- ISO 9001 Quality Management System, IIRSM Certified
- Greenhouse Production and Business Management Program, TOPIAN NEOM & Estidamah
- Life Cycle Assessment (LCA) and Sustainability Consulting, Coursera Certified
- Power BI and Microsoft Excel, Certified
- Member, Sustainability Professionals of Saudi Arabia, since May 2025
- Advanced Diploma in Wildlife Conservation, Alison
- Advanced AI Application Development, Tuwaiq Academy, 2026
- Certificate of Appreciation, Environment Fund, for active participation and contribution to developing innovative environmental solutions during the first Envirothon 2026
- Circular Economy and Resource Efficiency, National Academy for the Environment, 2026

LANGUAGES

Arabic: Native | **English:** IELTS 5.5, STEP 79